ECEN 743: Reinforcement Learning

Introduction

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What is Reinforcement Learning?

What is Machine Learning?

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- [Murphy, 2012]: "We define machine learning as a set of methods that can automatically detect patterns in data, and then use the uncovered patterns to predict future data, or to perform other kinds of decision making under uncertainty"

[Murphy, 2012] Kevin Murphy, "Machine Learning, A Probabilistic Perspective", MIT Press, 2012.

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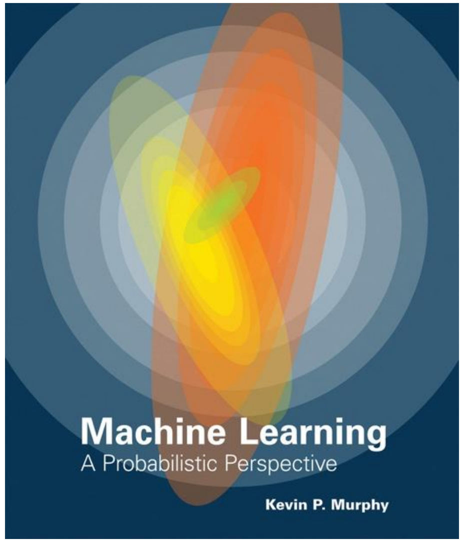

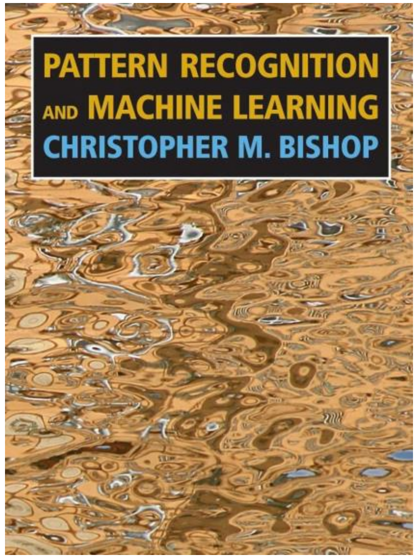

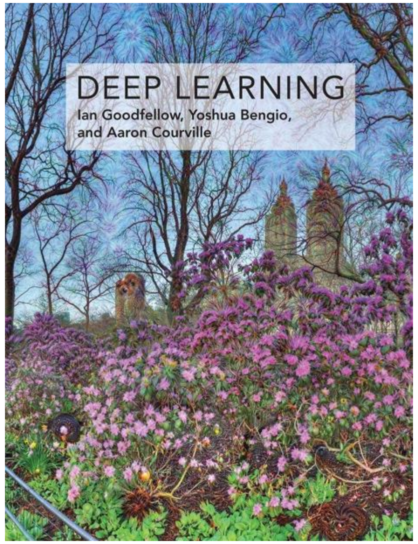

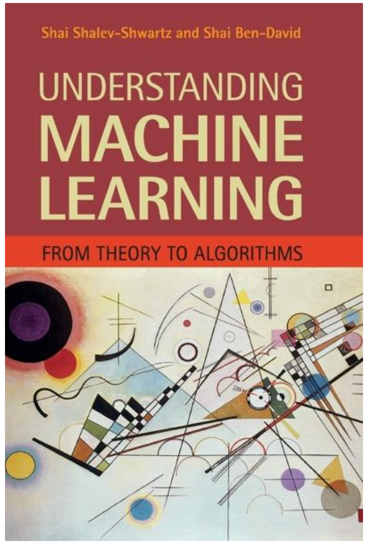

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Three Main Classes of Machine Learning

- Supervised learning
- Unsupervised learning
- Reinforcement learning

Supervised Learning

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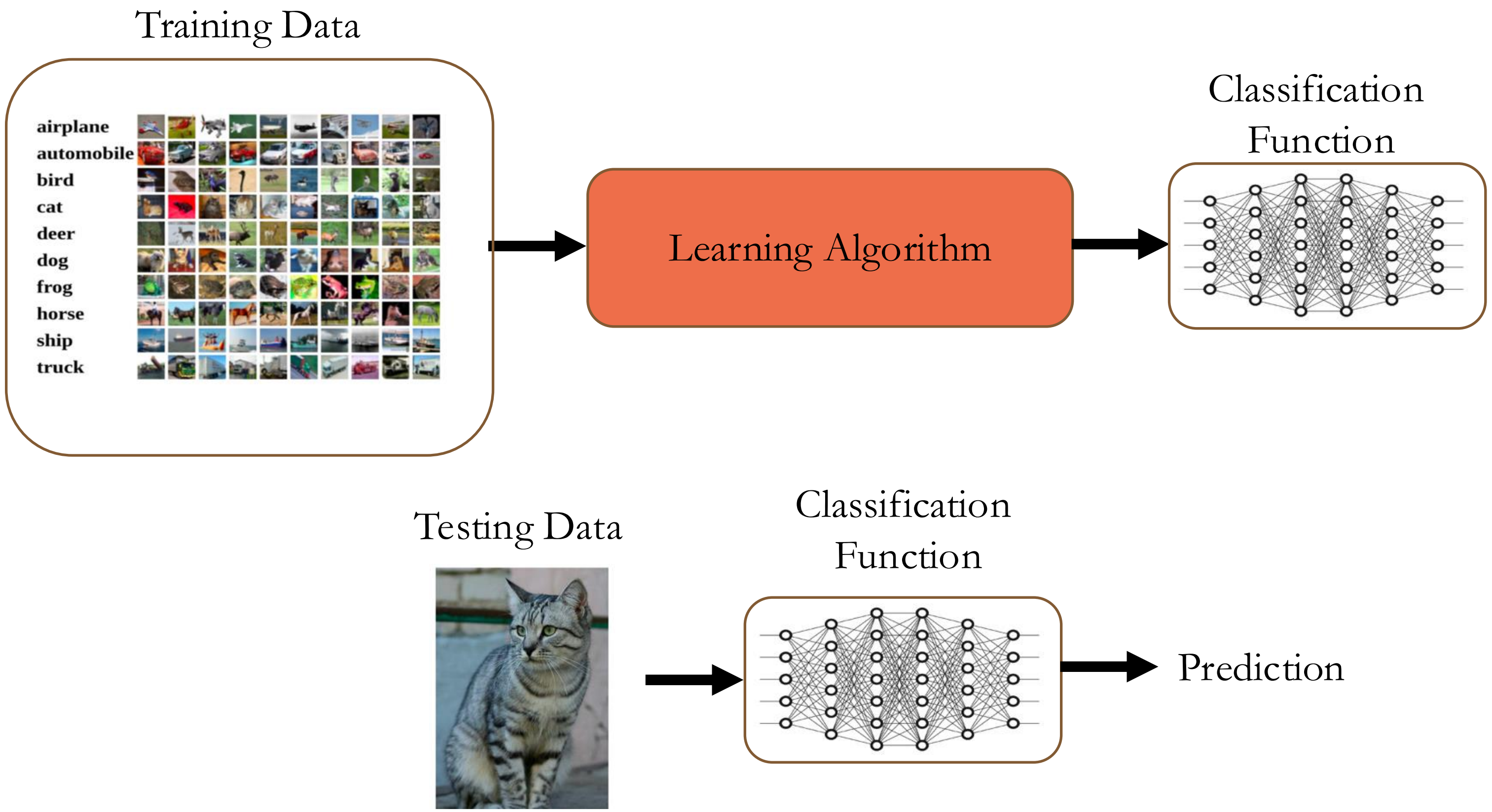

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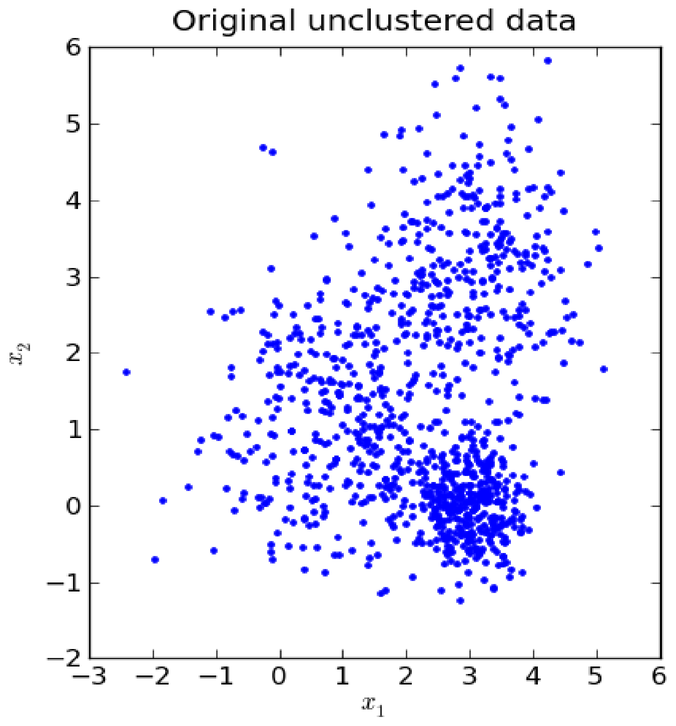

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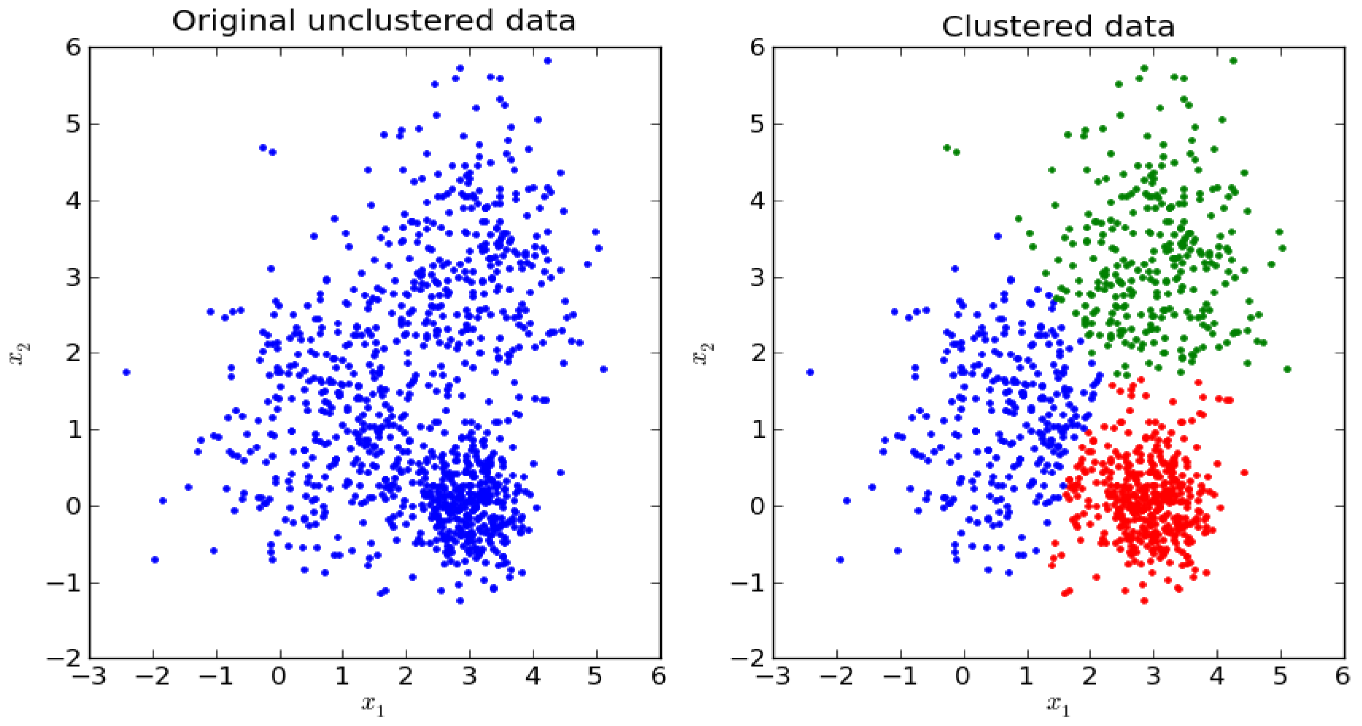

Reinforcement Learning

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- Agent and the environment:
  - At each time step t, the agent:
    - Gets an observation o_t
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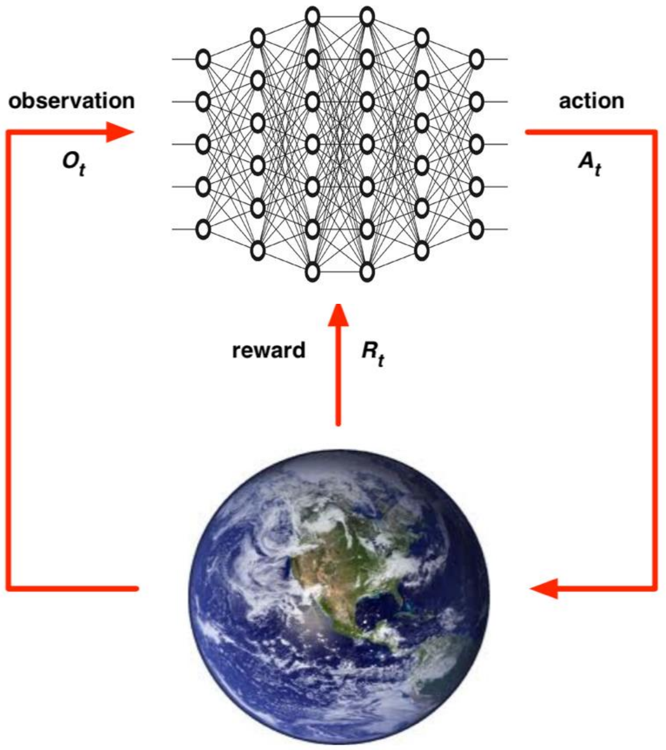
  - The goal is to select the sequence of actions in order to maximize the cumulative reward



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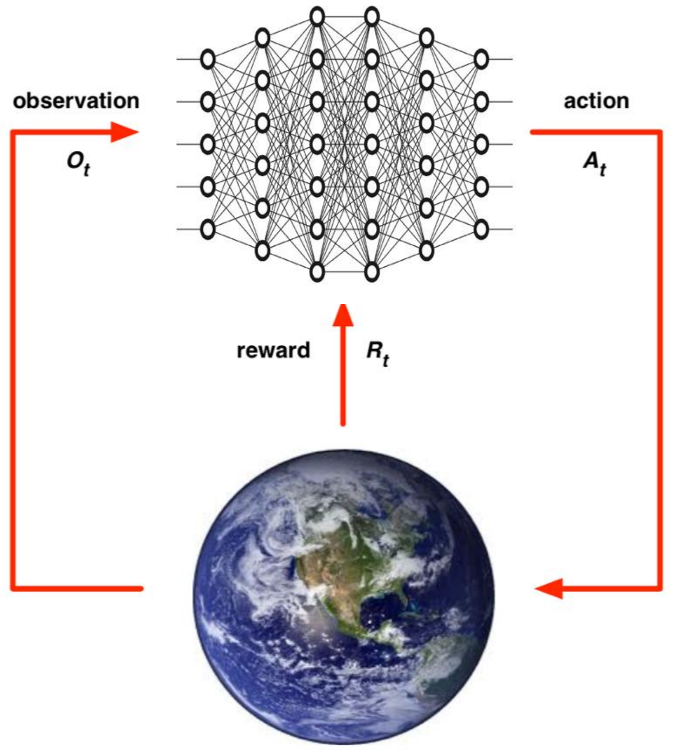

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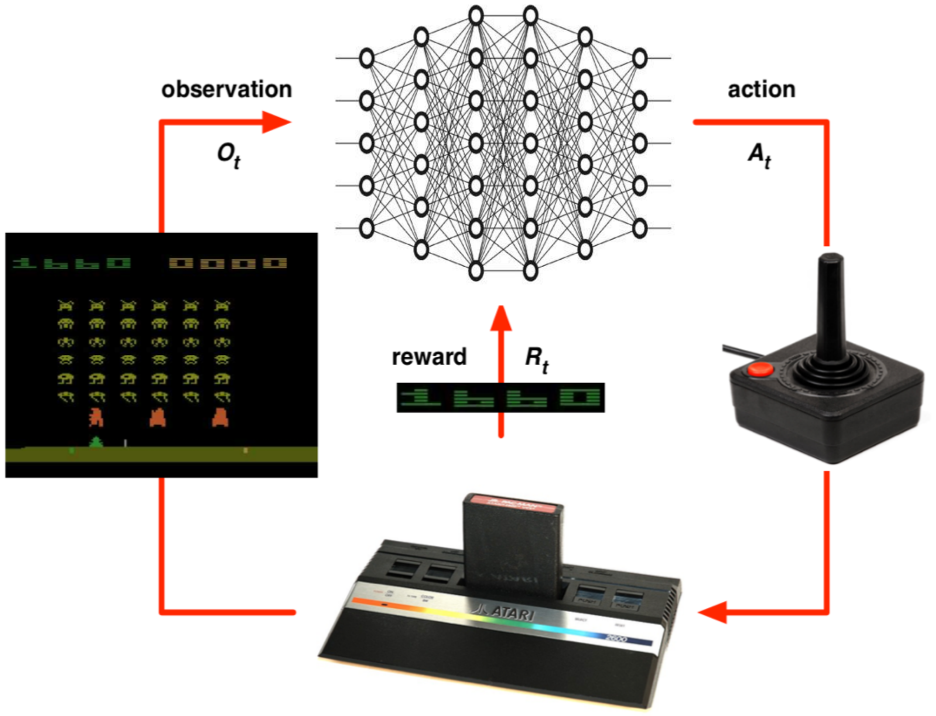
  - The model of the environment is unknown
    - Rules of the game are unknown

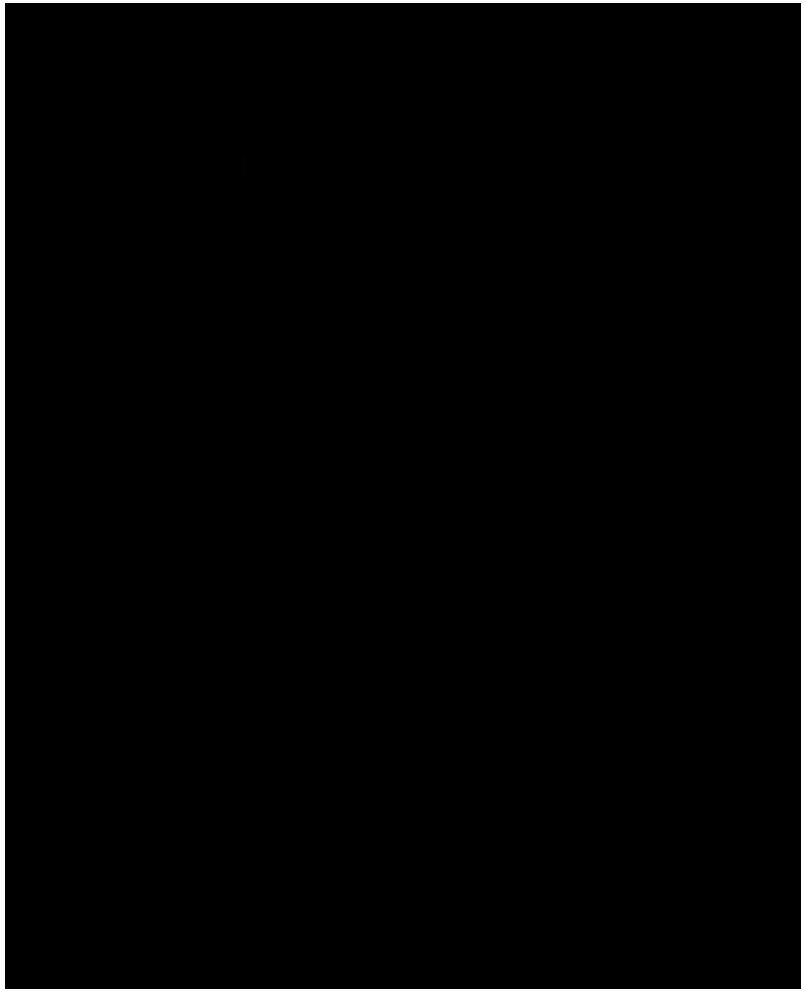


Reinforcement Learning: Challenges

- No prior data: RL agent (algorithm) generates data, by taking actions in an unknown environment
- Data generated is not i.i.d.
- No supervisor: optimal policy has to be learnt from the rewards observed in the self-generated data
- Actions have long term consequences, rewards are often delayed
- Dynamical systems issues: feedback and stability

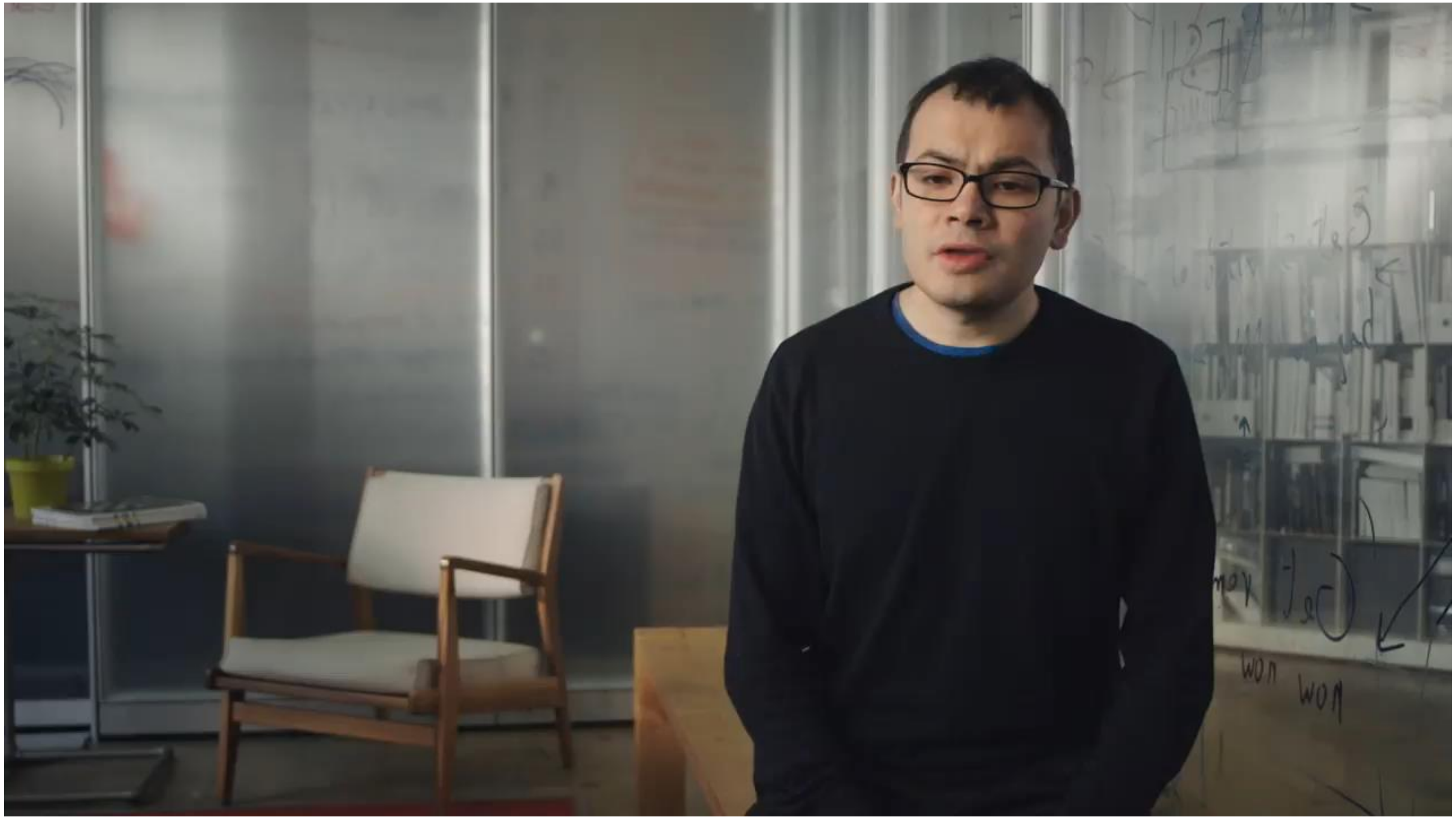
Reinforcement Learning: Examples

- RL for playing Atari Games (Mnih et al. "Human-level control through deep reinforcement learning", Nature, 2015)



Reinforcement Learning: Examples

- RL for playing Go (Silver et al. “Mastering the game of Go with deep neural networks and tree search”, Nature, 2016)



Reinforcement Learning: Examples

- RL for robotics (Kumar et al. “RMA: Rapid Motor Adaptation for Legged Robots”, RSS, 2021)

Rapid Motor Adaptation for Legged Robots

Ashish Kumar
UC Berkeley

Zipeng Fu
CMU

Deepak Pathak
CMU

Jitendra Malik
UC Berkeley/FAIR

Robotics: Science and Systems 2021

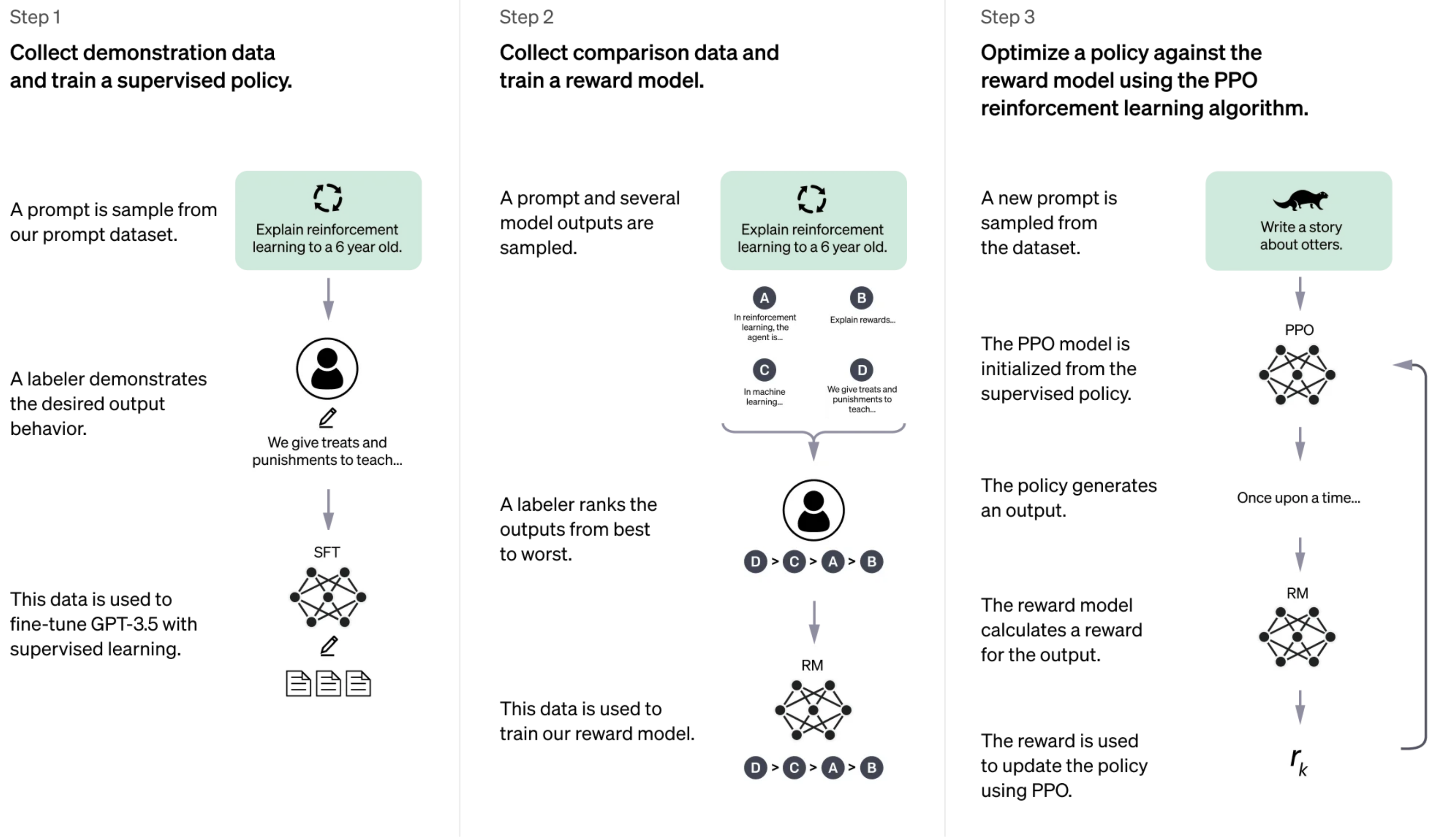
Reinforcement Learning: Examples

- RL for drone control (V. Saj, B. Lee, D. Kalathil, M. Benedict, "Robust Reinforcement Learning Algorithm for Vision-based Ship Landing of UAVs", 2022)

Reinforcement Learning: Examples

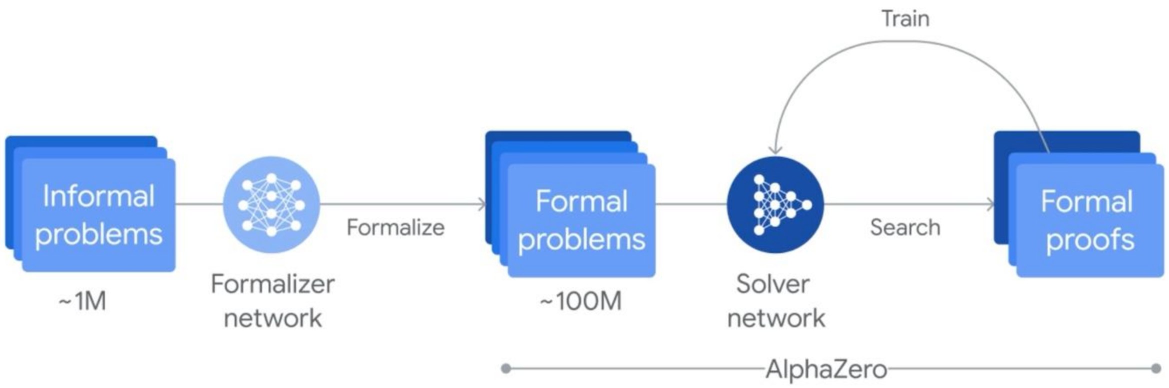
- GPT → InstructGPT → ChatGPT

Reinforcement Learning with Human Feedback (RLHF)



Reinforcement Learning: Examples

Google DeepMind About Research Technologies Discover



Process infographic of AlphaProof's reinforcement learning training loop: Around one million informal math problems are translated into a formal math language by a formalizer network. Then a solver network searches for proofs or disproofs of the problems, progressively training itself via the AlphaZero algorithm to solve more challenging problems.

RESEARCH

AI achieves silver-medal standard solving International Mathematical Olympiad problems

25 JULY 2024

AlphaProof and AlphaGeometry teams

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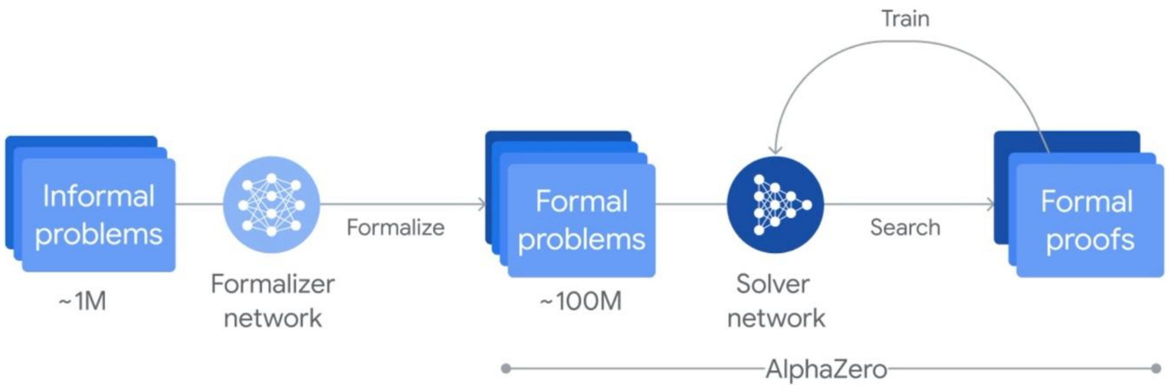

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Reinforcement Learning: Examples

OpenAI   Research   Products   Safety   Company

September 12, 2024

Learning to reason with LLMs

We are introducing OpenAI o1, a new large language model trained with reinforcement learning to perform complex reasoning. o1 thinks before it answers—it can produce a long internal chain of thought before responding to the user.

Contributions   Use o1 ↗

OpenAI o1 ranks in the 89th percentile on competitive programming questions (Codeforces), places among the top 500 students in the US in a qualifier for the USA Math Olympiad (AIME), and exceeds human PhD-level accuracy on a benchmark of physics, biology, and chemistry problems (GPQA). While the work needed to make this new model as easy to use as current models is still ongoing, we are releasing an early version of this model, OpenAI o1-preview, for immediate use in ChatGPT and to trusted API users.

Our large-scale reinforcement learning algorithm teaches the model how to think productively using its chain of thought in a highly data-efficient training process. We have found that the performance of o1 consistently improves with more reinforcement learning (train-time compute) and with more time spent thinking (test-time compute). The constraints on scaling this approach differ substantially from those of LLM pretraining, and we are continuing to investigate them.

Reinforcement Learning: Examples

deepseek

DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning

DeepSeek-AI

research@deepseek.com

Abstract

General reasoning represents a long-standing and formidable challenge in artificial intelligence. Recent breakthroughs, exemplified by large language models (LLMs) (Brown et al., 2020; OpenAI, 2023) and chain-of-thought prompting (Wei et al., 2022b), have achieved considerable success on foundational reasoning tasks. However, this success is heavily contingent upon extensive human-annotated demonstrations, and models' capabilities are still insufficient for more complex problems. Here we show that the reasoning abilities of LLMs can be incentivized through pure reinforcement learning (RL), obviating the need for human-labeled reasoning trajectories. The proposed RL framework facilitates the emergent development of advanced reasoning patterns, such as self-reflection, verification, and dynamic strategy adaptation. Consequently, the trained model achieves superior performance on verifiable tasks such as mathematics, coding competitions, and STEM fields, surpassing its counterparts trained via conventional supervised learning on human demonstrations. Moreover, the emergent reasoning patterns exhibited by these large-scale models can be systematically harnessed to guide and enhance the reasoning capabilities of smaller models.

The New York Times

Artificial Intelligence › | Data Centers in Space? | Killer A.I. Drones | New Billionaires | Debt Investors Grow Wary | U.S. Dismisses A.I. Risks

What to Know About DeepSeek and How It Is Upending A.I.

How did a little-known Chinese start-up cause the markets and U.S. tech giants to quake? Here's what to know.

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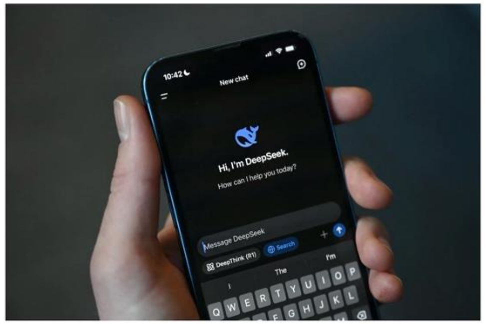

DeepSeek is a Chinese start-up run by a quantitative stock trading firm called High-Flyer. Greg Baker/Agence France-Presse — Getty Images

By Cade Metz
Reporting from San Francisco

Jan. 27, 2025
Leer en español

Tech stocks tumbled. Giant companies like Meta and Nvidia faced a barrage of questions about their future. Tech executives took to social media to proclaim their fears.

ARTIFICIAL INTELLIGENCE

Nvidia loses US$600 billion in a day after Chinese startup DeepSeek reveals a cheap AI model

Reinforcement Learning: Examples

- 2025 was the year of
  - Reasoning LLM
  - LLM Agent
  - Coding Agents
  - https://lmarena.ai/leaderboard https://www.vellum.ai/llm-leaderboard

- 2026 is going to be the of …

Reinforcement Learning: Examples

- YouTube video compression from DeepMind (Link)
- Several application from Microsoft, including in Azure platform, recommendation systems, video streaming, robotics (Link)
- Fast matrix multiplication algorithm from DeepMind (Link)
- Optimizing and finetuning ChatGPT (Link)
- Fast chip design algorithm from Google (Link)
- Optimizing recommendation systems from Netflix (Link)
- Optimizing recommendation systems from Google (Link)
- Playing strategic games from Meta (Link)

Course Syllabus Overview